

Revised Generation Interconnection System Impact Study Report

For Queue Project Q18-1106

Van Buren 138 kV
December 17, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1106. The Interconnection Customer proposes a 37.7 MW solar photovoltaic facility interconnecting at the Van Buren 138 kV substation in the Pennsylvania area. The request was received on November 18, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 281 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$285,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
W Pint – Powell 138 kV	West Hempfield – O'Fallon Renewable Energy Ctr 138 kV	109.0	290	31.0
Bge Timonium – Cipres 138 kV Line	Caney Crk – Mount Washington 138 kV	105.5	239	6.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Powell – W Pt 138 kV	Reconductor line	\$5,510,000	5% (\$273,000)
Bge Timonium – Cipres 138 kV	Reconductor line	\$956,000	1% (\$12,000)
	Total allocated to project		\$285,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.